

MGIS 355 Business Intelligence
Prof. Bui
Project #3

Instruction:

In this project, you will either work in a **team of three or four** to perform predictive analytics and compete in a Kaggle competition. The Kaggle competition for this assignment has been set up here: <https://www.kaggle.com/t/97f9e96690f54b37ae3bd47cb52fbdbd>

(Note: You can only join the competition using the link above. Once join, bookmark it so you can easily find it.)

Scenario:

Majestic Media (M²) is a telecom company in the Northeast area of the US. The company has experienced rapid growth in the last few years. Currently, the growth has slowed down, and the CEO is wondering how well the company will retain its customers. Among the many strategies, the CEO is particularly concerned with Majestic Media's customer churn rate.

Customer churn is the percentage of customers that stopped using your company's product or service during a certain time frame. Because it costs more to obtain new customers in such a saturated market, the CEO has made it a priority to reduce customer churn rate to improve the revenue for the company. The CEO has hired your team to advise on how to reduce the company's customer churn rate.

The company has released to your team a dataset of its customers in the last 12 months. Your objective in this project is to create a predictive model to classify whether a customer will terminate the service after 12 months (the churn).

Data Description:

Two datasets have been provided which contains information about the customer base of Majestic Media in the last 12 months. See Table below for description of variables included.

- The data set in the "Kaggle_churn_training.csv" file should be used for training your predictive models.
- The data sets in the "Kaggle_churn_test.csv" file should be used for the prediction.

The data dictionary is below:

Column name	Description	Date Type
Customer_ID	Unique customer ID	Numeric
Call_Failures	The number of call failures the customer experienced in the last 12 months	Numeric
Complains	Whether the customer has filed complains in the last 12 months	Categorical (0: No complaint, 1: complain)
Subscription_Length	Total months of subscription with the company	Numeric

Charge_Amount	The level of charge for the customer	Categorical (0: lowest amount, 9: highest amount)
Seconds_of_use	Total seconds of calls in the last 12 months	Numeric
Frequency_of_use	Total number of calls in the last 12 months	Numeric
Frequency_of_SMS	Total number of text messages in the last 12 months	Numeric
Distinct_called_numbers	Total number of distinct phone calls in the last 12 months	Numeric
Age_Group	The age group of the customer	Categorical (1: younger age, 5: older age)
Tariff_Plan	Type of contract	Categorical (1: Pay as you go, 2: contractual)
Status	The active status of the customer in the last month	Categorical (1: active, 2: non-active)
Customer_Value	The calculated value of customer (using the company proprietary model)	Numeric
Churn	Whether the customer terminate the service after 12 months	Categorical (1: churn, 0: non-churn)

Evaluation:

In your report, use three different approaches to evaluate your models:

- 1) **Using categorization accuracy (CA):** the models should also be assessed and compared based on their accuracy in prediction. This accuracy is generated through the confusion matrix.
- 2) **Using Kaggle submission score:** the solution model can be uploaded on Kaggle to compete against models developed by other students. The Kaggle system will evaluate and provide the score for the model performance.
- 3) **Using cost matrix:** the models should be assessed and compared based on their monetary value for Majestic Media. This can be calculated using the following cost matrix based on empirical observation from the Customer Service department.

		Predicted	
		0 (no churn)	1 (churn)
Actual	0 (no churn)	0	\$0
	1 (churn)	-\$50	-\$25

According to the cost matrix, every customer who left (churn) with no prediction (hence no intervention) from the Customer Service department will cost the company \$50. However, if the customer is accurately predicted to leave (churn), early interventions have a 50% chance of

retaining the customer. The average cost of losing such a customer is only \$25. For every other cases, no extra cost is incurred for the company.

Report Requirements:

We will use the following report format.

1. A cover page with names of team members and well as their Kaggle username
2. **Introduction:** describe the overall objective of the report
3. **Data Preparation:** describe any EDA steps you take, any steps you did to transform or prepare the data (which includes data partition)
4. **Data Analysis** (equivalent to Model Building): describe in detail each model that you have developed and tested. Provide information on *model*, *algorithm*, and *parameters* used in your analysis.
5. **Results** (equivalent to Model Comparison): compare developed models using different approaches:
 - a. Accuracy from the confusion matrix
 - b. Kaggle score from the leaderboard
 - c. Misclassification cost using cost matrix

Requirement #1: The number of models will match the number of members in the team. For example, if there are three team members, there should be three different models built by the team, and if there are four team members, there should be four different models.

Requirement #2: You must have a summary table for your results using the table below. Only reporting the best result for each model. For example, if you run multiple models using a decision tree algorithm, you only have to report the result for the best decision tree model you built. Do this for every different models you build (e.g., decision tree, random forest, neural network, ensemble).

#	Model	Accuracy	Kaggle score (rank)	Username	Cost
1	For example: Decision tree with hyperparameter tuning (cp=0.02), 5-fold cross- validation Churn = Customer_Value + Age_Group	0.8334	0.8012 (12)	Kaggle- username	-\$1,250
2	Random forest with hyperparameter tuning, 5-fold cross-validation Churn = Customer_Value + Age_Group	0.8642	0.8354 (10)	Kaggle- username	-\$750

6. Conclusion:

- Provide your final conclusion on the model based on results from your evaluation. For example, “We recommend using a X model to predict fraud. The model is $\text{Churn} = \text{feature 1} + \text{feature 2} + \text{feature 3}$. The model will use X1 algorithm with the following parameters. This model has an accuracy of %%, a Kaggle score of ###, and a monetary value of \$\$\$\$.”
- Based on what you learn from your models, provide suggestions to the company to reduce the customer churn rate. (That is, your suggestions should relate to your models.)

Peer Evaluation Survey:

Each student is required to complete the peer evaluation survey (https://rit.az1.qualtrics.com/jfe/form/SV_eQCaZp7uYOxkokK) Failure to do so will result in penalty (see below).

Project 3 Grading Rubric:

- Kaggle submission scores** (20 pts): this measures the technical sophistication of your solutions. Teams will be ranked against each other using Kaggle submission systems. The distribution of points will be as follows:
 - Rank 1: 20 points
 - Rank 2-3: 17 points
 - Rank 4-5: 14 points
 - Rank 6-7: 11 points
 - Rank 8-9: 8 points
- Final report quality** (70 pts): Each report will be graded using the following rubric:

Criterion	Unsatisfactory	Satisfactory	Exemplary
	1	2	3
<i>Technical accuracy (15 pts)</i>	Multiple inaccuracies were identified in the presentation of findings.	Some minor inaccuracies were identified in the presentation of findings.	Analytical findings accurately represent the given data; no inaccuracies were identified.
<i>Report coherency (15 pts)</i>	Significant weaknesses in writing quality are identified; multiple typographical and grammatical errors are present.	Writing quality is generally strong; few typographical and grammatical errors are present.	The report is well-written and free of typographical and grammatical errors.
<i>Narrative flow (10 pts)</i>	Organization of the submission is difficult to follow; evidence of plagiarism exists.	Organization of submission is generally followable, but some re-organization may be needed.	All portions of the submission are well-organized and coherently presented.

<i>Professionalism (10 pts)</i>	Professional appearance of the submission is relatively low	Submissions are generally well-polished; some revision of aesthetic appearance or presentation would be beneficial	Submission reflects professional polish (e.g., pages numbered, all graphics appropriately referenced)
<i>Persuasiveness (20 pts)</i>	Recommendations/findings are not well oriented to the intended audience; persuasiveness of findings is limited	Recommendations are relatively persuasive; some revisions are needed	Recommendations are relatable to the intended audience and deemed insightful and persuasive

- **Team contributions** (10 pts): Each student is assumed to have full 10 points on team contribution. Students will lose points if:
 - a. The student fails to complete the Peer Evaluation Survey by the project deadline (https://rit.az1.qualtrics.com/jfe/form/SV_eQCaZp7uYOxkokK)
 - b. Other teammates agree that the student did not contribute equally to the project work. In this case, based on the teammates' comments appropriate points will be deducted from the student.

Submission requirements:

- Each team will submit a final report and all related files in MyCourses. The files can be submitted by one student on behalf of the team.
- Each team member needs to complete the Peer Evaluation Survey (https://rit.az1.qualtrics.com/jfe/form/SV_eQCaZp7uYOxkokK)